THERMAL EQUILIBRIUM: Two objects are said to be in thermal equilibrium, if they won't exchange any energy at macroscopic level if placed in contact. When this happens, we also say that the two bodies are at same temperature.

[When this happens, typically average kinetic energy per degree of freedom becomes same for the two bodies. This can be understood statistically, we will study more on it when we learn entropy].

Thus temperature is in a way measure of energy possessed by substance particles per degree of freedom.]

A hotter body has more energy per degree of freedom than a colder body and they keep exchanging energy until the energy per degree of freedom is equalized.

ZEROTH LAW OF THERMODYNAMICS: If two bodies A and B are in thermal equilibrium and A and C are also in thermal equilibrium then B and C are also in thermal equilibrium.

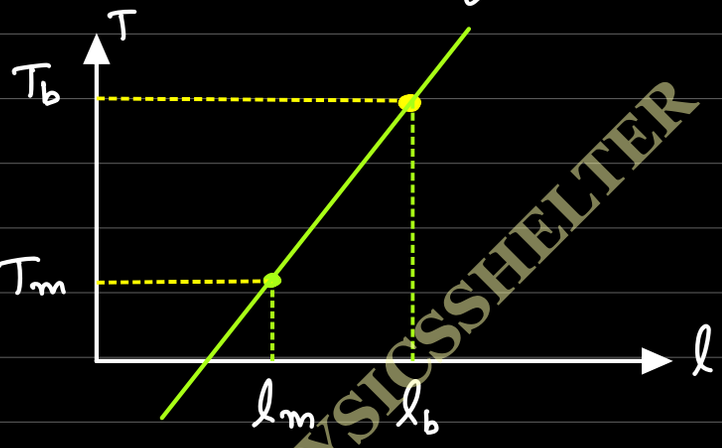
Zeroth law of thermodynamics forms the basis of measurement of temperature. We could choose some physical property which depends on energy per degree of freedom (say thermal expansion of mercury, kept in a glass tube). Now if the length of mercury column is same upon keeping it in contact with different bodies then the two bodies must be at same temperature which could be specified in units of length of mercury column.

DEFINING TEMPERATURE SCALES: To give numerical value to temperature we can choose property of a substance which changes monotonically with supplying heat to the substance and note its value at some standardized thermodynamic states (such as melting and boiling points of water). We can assign certain arbitrary values to the temperature at these states and then make a linear fit between temperature and the measured substance property to read temperature against the substance property.

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For instance let us assign a temperature value T_m to the length of mercury column (say l_m) when it is in thermal equilibrium with melting ice and similarly let us define a temperature value of T_b when the column is in equilibrium with boiling water. Let the column length now be l_b. Thus now we have two data points (l_m, T_m) and (l_b, T_b). Now we can fit a straight line between two points to read temperature against any length of column.





Now $$\frac{T - T_m}{l - l_m} = \frac{T_b - T_m}{l_b - l_m} \Rightarrow \boxed{T = T_m + \left(\frac{T_b - T_m}{l_b - l_m}\right)(l - l_m)} \quad (1)$$

Note that this definition is completely arbitrary because we have infinite choice for T_b and T_m and also choice of property varying with temperature. However certain scales are more popular than other scales. (For that matter we could also choose melting and boiling points of sodium.

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Celcius Scale (Centigrade Scale): In this scale, $T_m = 0^\circ C$ and $T_b = 100^\circ C$

$$\Rightarrow {}^\circ C = \left(\frac{l - l_m}{l_b - l_m}\right) \times 100 \quad (2)$$

Fahrenheit Scale: In this scale $T_m = 32^\circ F$, $T_b = 212^\circ F$

$\Rightarrow$ Using Equation (1): $$^\circ F = 32 + 180 \times \left(\frac{l - l_m}{l_b - l_m}\right) \quad (3)$$

Equations ② & ③ also enable us to convert temperature from °F to °C and vice versa. Eliminating $\left(\frac{l-l_m}{l_b-l_m}\right)$ from ② & ③:

$$\frac{°F-32}{180} = \frac{°C-0}{100} \Rightarrow \boxed{°F = 32 + \frac{9}{5}°C} \quad (4)$$

RESISTANCE THERMOMETER: In this type of thermometer we use the fact that electrical resistance of metals changes with temperature. Thus we could find the resistance of a wire at 0°C and 100°C and then fit a linear curve between two points thus obtained. From equation ① we can readily write:

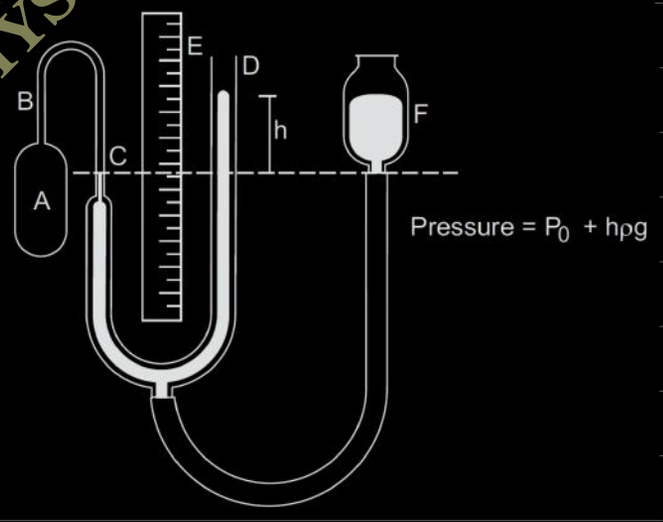
$$T(°P) = 0 + \left(\frac{R-R_m}{R_b-R_m}\right) \times 100°P \quad (5)$$

Constant Volume Thermometer:

In this thermometer we use the fact that pressure of a gas increases upon heating at a constant volume.

According to equation ①, in this standardization we could define temperature as:

$$T(°C) = \frac{(P-P_m)}{(P_b-P_m)} \times 100 \quad (6)$$



Here volume is maintained constant by increasing mercury level in a tube as shown in the figure above.

EXISTENCE OF MINIMUM POSSIBLE TEMPERATURE: At one point of time it was thought that temperature could be decreased indefinitely, however as statistical understanding of energy of matter particles at microscopic level developed, it was understood that if molecules were to stop colliding against

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walls of a vessel (Zero Kinetic Energy) then pressure would become zero. Thus we could evaluate a temperature at which pressure of an ideal gas becomes zero by setting $P=0$ in equation ①.

$\Rightarrow$ T for zero pressure $= \left(\frac{-P_m}{P_b - P_m}\right) \times 100^\circ C$. When this is calculated it comes out about $-273.16^\circ C$. Thus temperatures less than $-273.16^\circ C$ can't exist and this is therefore theoretically minimum achievable temperature.

KELVIN SCALE: In this scale we assign a zero numerical value to a temperature when ideal gas exerts no pressure and a numerical value of 273.16 to triple point of water, thus we have two data points and according to equation ①,

$$T(K) = 0 + \frac{P}{P_{tp}} \times 273.16 \quad K.$$

$\Rightarrow$ $\boxed{T = \frac{P}{P_{tp}} \times 273.16\ K}$ ↳ Triple Point Pressure.

Example: The pressure of air in bulb of a constant volume thermometer is 73 cm of mercury at $0^\circ C$, 100.3 cm at $100^\circ C$ and 77.8 cm of mercury at room temperature. Find the room temperature in centigrade.

Sol: According to equation ①, $T = T_m + \left(\frac{P - P_m}{P_b - P_m}\right) \times (T_b - T_m)$

$\Rightarrow$ $T = 0 + \frac{77.8 - 73}{100.3 - 73} \times 100 = 17.6^\circ C$

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Example:

The resistances of a platinum resistance thermometer at the ice point, the steam point and the boiling point of sulphur are 2·50, 3·50 and 6·50 Ω respectively. Find the boiling point of sulphur on the platinum scale. The ice point and the steam point measure 0° and 100° respectively.

Sol: $T = T_m + \left(\frac{R - R_m}{R_b - R_m}\right) \times (T_b - T_m) = 0 + \frac{6.5 - 2.5}{3.5 - 2.5} \times 100$

$\Rightarrow T = 400^\circ$ (Platinum Scale).

→ Note that just because ice point and steam point are given same values as for a mercury thermometer, it does not ensure at all temperatures because, different monotonic functions between same starting and end points may have different shapes hence the word "Platinum scale".

Example:

A platinum resistance thermometer reads 0° and 100° at the ice point and the boiling point of water respectively. The resistance of a platinum wire varies with Celsius temperature θ as $R_t = R_0 (1 + \alpha\theta + \beta\theta^2)$, where $\alpha = 3{\cdot}8 \times 10^{-3}\,^\circ\mathrm{C}^{-1}$ and $\beta = -5{\cdot}6 \times 10^{-7}\,^\circ\mathrm{C}^{-2}$. What will be the reading of this thermometer if it is placed in a liquid bath maintained at 50°C ?

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Solution: $T(^\circ P) = 0 + \left(\frac{R - R_m}{R_b - R_m}\right) \times 100$

Now $R_m = R(0) = R_0[1 + \alpha \times 0 + \beta \times 0) = R_0$

$R_b = R_0[1 + \alpha \times 100 + \beta \times 100^2) = R_0(1 + 0.38 - 0.0056)$

$\Rightarrow R_b = R_0(1.3744)$, $R = R_0[1 + \alpha \times 50 + \beta \times 50^2]$

$\Rightarrow R = R_0[1 + 0.19 - 0.0014] = R_0(1.1886)$

$\Rightarrow T(^\circ P) = \left(\frac{1.1886 - 1}{1.3744 - 1}\right) \times 100 = 50.4\ ^\circ P$

→ This example clearly illustrates how platinum and mercury thermometers would disagree in general even though they will agree at ice point and boiling points.

Example:

A platinum resistance thermometer is constructed which reads 0° at ice point and 100° at steam point. Let t_p denote the temperature on this scale and let t denote the temperature on a mercury thermometer scale. The resistance of the platinum coil varies with t as $R_t = R_0(1 + \alpha t + \beta t^2)$. Derive an expression for the resistance as a function of t_p.

Solution: $t_p = \frac{R - R_0}{R_0(1 + 100\alpha + \beta \times 100^2) - R_0} \times 100$

$\Rightarrow t_p = \frac{R - R_0}{100(\alpha + 100\beta)R_0} \times 100 \Rightarrow R = R_0 t_p(\alpha + 100\beta) + R_0$

$\Rightarrow R = R_0[1 + \alpha t_p + 100\beta t_p]$ Ans.

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REAUMEAUR SCALE: In this scale, ice point is taken as 0° and boiling point is taken as 80°.

Example: Find conversion formula between Reaumur Scale ($^\circ R$) and Celsius Scale).

Solution: $$^\circ R = \frac{l - l_m}{l_b - l_m} \times 80 \quad , \quad ^\circ C = \frac{l - l_m}{l_b - l_m} \times 100$$

$$\Rightarrow \quad ^\circ R = \frac{4}{5}\,^\circ C$$

Example: Find Conversion formula between $^\circ F$ and $^\circ R$.

Solution: $$^\circ F = 32 + \frac{l - l_m}{l_b - l_m} \times 180 \quad , \quad ^\circ R = \frac{l - l_m}{l_b - l_m} \times 80$$

From two equations: $^\circ F = 32 + {}^\circ R \times 9/4$ dms.

Example: At what temperature will numerical values of Reaumur scale and Fahrenheit scale will be same?

Sol: Let the numerical value be $x \Rightarrow x = 32 + \frac{9}{4}x$

$$\Rightarrow \quad \frac{5}{4}x = -32 \quad \Rightarrow x = -25.6 \quad \Rightarrow -25.6^\circ F = -25.6\,^\circ R.$$

Example: In a temperature scale X, ice point and boiling point are respectively taken as $20^\circ X$ and $220^\circ X$, while in another scale Y, these are taken as $-20^\circ Y$ and $380^\circ Y$. At what temperature is the numerical value same in both scales.

Sol: $$^\circ X = 20 + \left(\frac{l - l_m}{l_b - l_m}\right) \times 200, \quad ^\circ Y = -20 + \left(\frac{l - l_m}{l_b - l_m}\right) \times 400$$

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$\Rightarrow \quad {}^\circ X = 20 + \left(\frac{{}^\circ Y + 20}{2}\right) = 30 + {}^\circ Y/2$

Let the two read same numerical value x for some temp.

$\Rightarrow \quad x = 30 + \frac{x}{2} \quad \Rightarrow \quad x = 60 \quad \Rightarrow \quad 60^\circ X = 60^\circ Y$ Ans.

ABSOLUTE SCALES: An absolute scale is one which reads numerically zero at 0K. [When gas pressure is zero].

EXAMPLE: Two absolute scales X and Y have triple point of water defined as $300^\circ X$ and $450^\circ Y$ respectively. How are T_X and T_Y related?

Solution: $\quad {}^\circ X = 0 + \frac{P}{P_{tr}} \times 300 \quad , \quad {}^\circ Y = \frac{P}{P_{tr}} \times 450$

$\Rightarrow \quad {}^\circ X = \frac{2}{3}\,{}^\circ Y$ Ans.

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